

Retrieve number and ratio

Use the scaffold, show each step and ask for an adult check when marked.

Name: _____ Date: _____

1. Read the model: Calculate $2 \frac{1}{4}$ divided by 3, then write the answer as a decimal. Copy or complete the first step.

2. Solve $5x + 8 = 73$ and convert 3.45 km to metres.

3. Find the area of a triangle with base 18 cm and height 11 cm, then find the mean of 18, 22, 25 and 15.

4. Miro says: Miro records a general target such as 'get better at maths'. Is this correct? Explain.

Teacher answers and checking prompts

1. Read the model: Calculate $2\frac{1}{4}$ divided by 3, then write the answer as a decimal. Copy or complete the first step.

Answer $\frac{3}{4}$ or 0.75.

2. Solve $5x + 8 = 73$ and convert 3.45 km to metres.

Answer $x = 13$ and 3,450 m.

3. Find the area of a triangle with base 18 cm and height 11 cm, then find the mean of 18, 22, 25 and 15.

Answer 99 cm² and 20.

4. Miro says: Miro records a general target such as 'get better at maths'. Is this correct? Explain.

Answer Use assessment evidence to name the exact concept, method, example and support required next.

Retrieve number and ratio

Work independently. Show a complete method and check at least one answer.

Name: _____ Date: _____

1. Calculate $2\frac{1}{4}$ divided by 3, then write the answer as a decimal.

6. Check this result using an inverse, estimate or known fact: Find the area of a triangle with base 18 cm and height 11 cm, then find the mean of 18, 22, 25 and 15.

2. Solve $5x + 8 = 73$ and convert 3.45 km to metres.

7. Prove or explain your reasoning: A £640 item is reduced by 12.5%. Find the sale price.

3. Find the area of a triangle with base 18 cm and height 11 cm, then find the mean of 18, 22, 25 and 15.

8. Identify and correct this misconception: Miro records a general target such as 'get better at maths'.

4. Solve this independently and show every stage: Calculate $2\frac{1}{4}$ divided by 3, then write the answer as a decimal.

9. Write a complete sentence explaining how you checked one answer.

5. Use a second method or representation for: Solve $5x + 8 = 73$ and convert 3.45 km to metres.

10. Write and solve a similar question to: Find the area of a triangle with base 18 cm and height 11 cm, then find the mean of 18, 22, 25 and 15.

Teacher answers and checking prompts

1. Calculate $2\frac{1}{4}$ divided by 3, then write the answer as a decimal.
Answer $\frac{3}{4}$ or 0.75.
2. Solve $5x + 8 = 73$ and convert 3.45 km to metres.
Answer $x = 13$ and 3,450 m.
3. Find the area of a triangle with base 18 cm and height 11 cm, then find the mean of 18, 22, 25 and 15.
Answer 99 cm² and 20.
4. Solve this independently and show every stage: Calculate $2\frac{1}{4}$ divided by 3, then write the answer as a decimal.
Answer $\frac{3}{4}$ or 0.75.
5. Use a second method or representation for: Solve $5x + 8 = 73$ and convert 3.45 km to metres.
Answer Expected result: $x = 13$ and 3,450 m. Method or representation may vary.
6. Check this result using an inverse, estimate or known fact: Find the area of a triangle with base 18 cm and height 11 cm, then find the mean of 18, 22, 25 and 15.
Answer Expected result: 99 cm² and 20. A valid check must be shown.
7. Prove or explain your reasoning: A £640 item is reduced by 12.5%. Find the sale price.
Answer £560. The explanation must show why the method works.
8. Identify and correct this misconception: Miro records a general target such as 'get better at maths'.
Answer Use assessment evidence to name the exact concept, method, example and support required next.
9. Write a complete sentence explaining how you checked one answer.
Answer Answer should name an inverse, estimate, boundary, known fact or equivalent representation.
10. Write and solve a similar question to: Find the area of a triangle with base 18 cm and height 11 cm, then find the mean of 18, 22, 25 and 15.
Answer Answers vary. The new question must use the same mathematical structure and include a correct solution.

Retrieve number and ratio

Prove, compare or generalise. Write enough reasoning for another pupil to follow.

Name: _____ Date: _____

1. Prove or explain your reasoning: A £640 item is reduced by 12.5%. Find the sale price.

6. Create a new example that tests the same idea as: Find the area of a triangle with base 18 cm and height 11 cm, then find the mean of 18, 22, 25 and 15.

2. Prove the answer to this question, rather than only calculating it: Calculate $2\frac{1}{4}$ divided by 3, then write the answer as a decimal.

7. Find a second method or representation. Compare its efficiency with your first method.

3. Solve this in two different ways and compare them: Solve $5x + 8 = 73$ and convert 3.45 km to metres.

8. Write one always, sometimes or never statement about today's learning and justify it.

4. Work backwards from the answer to reconstruct the question: 99 cm² and 20.

9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.

5. Explain why this reasoning fails: Miro records a general target such as 'get better at maths'.

10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.

Teacher answers and checking prompts

1. Prove or explain your reasoning: A £640 item is reduced by 12.5%. Find the sale price.
Answer £560. The explanation must show why the method works.
2. Prove the answer to this question, rather than only calculating it: Calculate $2\frac{1}{4}$ divided by 3, then write the answer as a decimal.
Answer Expected result: $\frac{3}{4}$ or 0.75. Proof or justification must match the lesson structure.
3. Solve this in two different ways and compare them: Solve $5x + 8 = 73$ and convert 3.45 km to metres.
Answer Expected result: $x = 13$ and 3,450 m. Two valid methods and a comparison are required.
4. Work backwards from the answer to reconstruct the question: 99 cm² and 20.
Answer One valid reconstruction is: Find the area of a triangle with base 18 cm and height 11 cm, then find the mean of 18, 22, 25 and 15. Other equivalent questions are acceptable.
5. Explain why this reasoning fails: Miro records a general target such as 'get better at maths'.
Answer Use assessment evidence to name the exact concept, method, example and support required next.
6. Create a new example that tests the same idea as: Find the area of a triangle with base 18 cm and height 11 cm, then find the mean of 18, 22, 25 and 15.
Answer Answers vary. The example and solution must preserve the mathematical structure.
7. Find a second method or representation. Compare its efficiency with your first method.
Answer Answers vary. Comparison should refer to accuracy, number structure and clarity.
8. Write one always, sometimes or never statement about today's learning and justify it.
Answer Answers vary. A valid example and counterexample should support the classification.
9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.
Answer Answers vary. The error must be mathematically relevant and the correction must be precise.
10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.
Answer Answers vary. The explanation must distinguish the mathematical structure from the changed value.